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NETWORK PROGRAMMING · TOPIC 1

Online Chess Application

Players log in, find an opponent (randomly, by skill level or by invitation) and play chess with each other over the Internet.

Team

Doãn Khang · [Member 2] · [Member 3]

ONE LOVE. ONE FUTURE.

Main features

1. Account & lobby

- Log in with an account
- See who is online
- At login, choose how to be matched: random or by skill level

2. Finding an opponent

- Random match with other random-mode players
- Skill match with players of similar rating
- Create a room, invite a player, accept an invitation

3. Playing a game

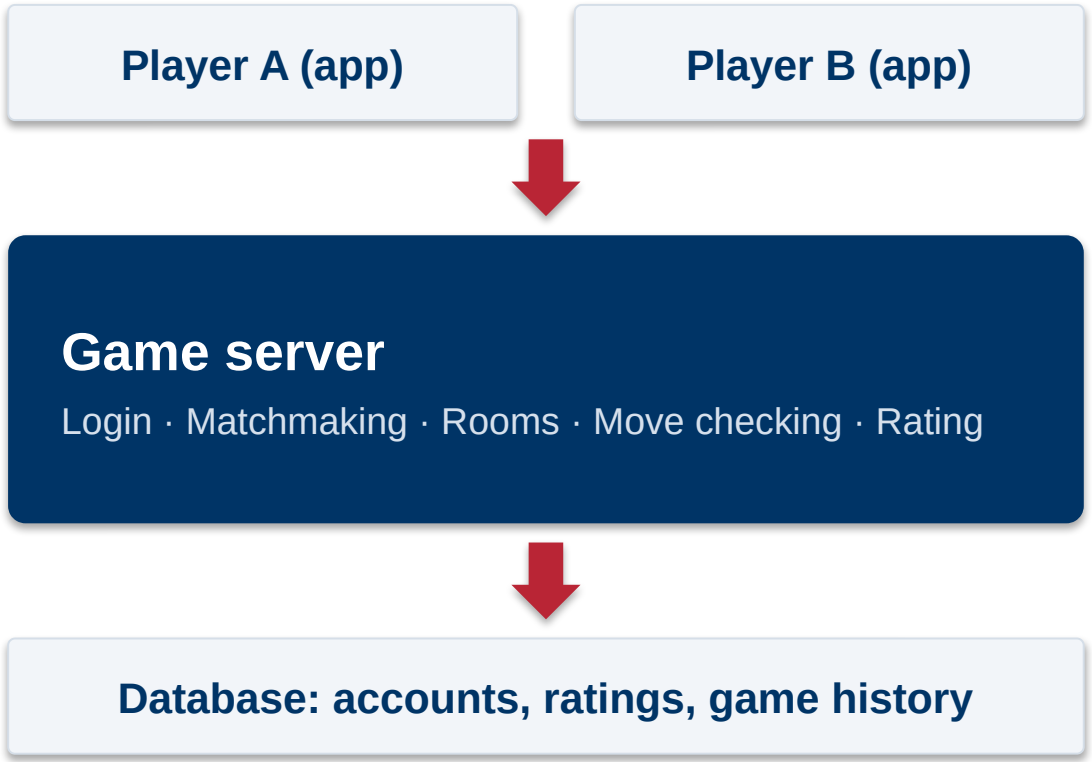
- Players take turns; the server checks every move is legal
- The server announces the winner and loser to both
- Offer a draw; the opponent accepts or declines

4. Rating & ranking

- The server updates each player's rating after every game
- See the leaderboard sorted by rating

Architecture & approach

Client – server model over TCP



The server is the referee

Every move is sent to the server to be checked. The server decides who wins, so players cannot cheat.

Elo rating

Beating a stronger player earns more points; losing to a weaker one costs more. Ratings drive skill matching and the leaderboard.

Planned technology

Java (TCP sockets, multithreaded server) · Swing user interface · MySQL database

Challenges

1. Many players at the same time

The server handles many clients at once, and they share the online list, queues and rooms.

Plan: one thread per client, and lock shared data so two threads never change it at the same time.

2. A clear protocol

TCP sends a stream of bytes, not separate messages, and a client may send bad or out-of-turn data.

Plan: one message per line with a fixed format; the server rejects invalid or out-of-turn messages.

3. Players disconnecting

A player may lose connection or close the app in the middle of a game.

Plan: the server waits a short time for them to come back, then counts it as a loss.

4. Full chess rules

Castling, en passant, promotion, checkmate and stalemate are easy to get wrong.

Plan: use a tested chess library (if allowed) or write a test for each rule.